

Public Economics (ECON 131)

Section #7: Capital Income and Savings Taxation

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March 9, 2025

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1 Capital Income and Savings Taxation

1.1 Key concepts

- Intertemporal choice model: The choice individuals make about how to allocate their consumption over time. before: one period now: two periods
- Savings: The excess of current income over current consumption.
- Intertemporal budget constraint (IBC): The measure of the rate at which individuals can trade off consumption in one period for consumption in another period. Note, [the opportunity cost of first-period consumption is the interest income not earned on savings for second-period.]
- We find the optimal consumption level at each time by solving a lifetime utility maximization problem.

1.2 Intertemporal Choice, Graphically

Using the information from the IBC we can graph the opportunity set. A tax on savings modifies the individual's intertemporal choice. This change can be decomposed in two effects, the substitution and the income effect:

	income	consumption
Period 1	Y	C_1
Period 2	0	C_2

⇒ you are going to save money earned in period 1 to consume in period 2. so, how much do you save? $S = Y - C_1$
(whatever you don't consume in period 1)

⇒ We earn interest (r) on savings, s.t. the amount you consume in period 2 is $C_2 = S(1+r)$

↗ want in terms of $C_1 + C_2$
(IBC)

intertemporal budget constraint where no tax

$$Y = C_1 + S = C_1 + \frac{C_2}{(1+r)}$$

if $C_1 = 0$, $Y = \frac{C_2}{(1+r)} \Rightarrow C_2 = Y(1+r)$

if $C_2 = 0$, $Y = C_1$

a tax on savings can happen (only tax gains on savings)

$$C_2 = S(1+r(1-\tau))$$

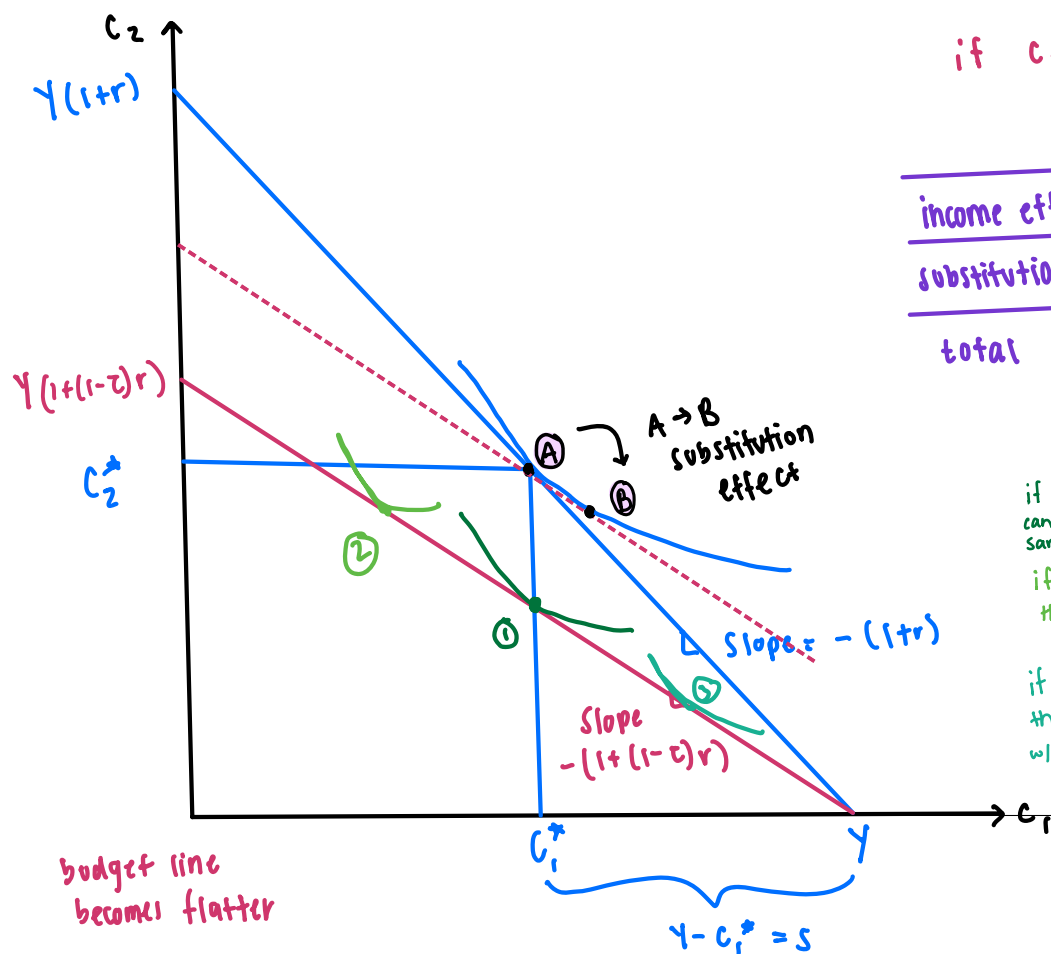
IBC where tax on savings

$$Y = C_1 + S = C_1 + \frac{C_2}{(1+r(1-\tau))}$$

if $C_1 = 0$, $Y = \frac{C_2}{(1+r(1-\tau))}$

$$\Rightarrow C_2 = Y(1+r(1-\tau))$$

if $C_2 = 0$, $Y = C_1$



	C_1	C_2
income effect	↓	↓
substitution	↑	↓
total	?	↓

(levels)
Feel poorer!

slopes

↓
before, consuming 1 unit in period 2 requires giving up $(1+r)$ units of consumption in period 1.

now, consuming 1 unit in period 2 requires giving up $(1+(1-\tau)r)$ units of consumption in period 2.

if @ B, income and sub effect cancel each other out. C_1 is the same w/ and w/o tax

if @ C, income effect dominates the sub effect. C_1 lower w/ tax than w/o tax

if @ A, sub effect dominates the income effect. C_1 higher w/ tax than w/o tax

consumption in period 2 has become relatively cheaper

1.3 Practice problem

1.3.1 Gruber, Ch. 22, Q.13

Consider a model in which individuals live for two periods and have utility functions of the form $U(C_1, C_2) = \ln C_1 + \ln C_2$. They earn income of \$100 in the first period and save S to finance consumption in the second period. The interest rate, r , is 10%.

- (a) Set up the individual's lifetime utility maximization problem. Solve for the optimal C_1 , C_2 , and S . (Hint: Rewrite C_2 in terms of income, C_1 , and r .) Draw a graph showing the opportunity set.

$$\text{IBC: } Y = C_1 + \frac{C_2}{(1+r)}$$

$$\Rightarrow 100 = C_1 + \frac{C_2}{(1+0.1)}$$

$$\Rightarrow 100 = C_1 + \frac{C_2}{1.1}$$

$$\Rightarrow (100 - C_1) = \frac{C_2}{1.1}$$

$$\Rightarrow C_2 = (100 - C_1)(1.1) \checkmark$$

$$\left[\begin{array}{l} \text{alternatively, we know} \\ C_2 = S(1+r) = (Y - C_1)(1+r) \\ = (100 - C_1)(1.1) \checkmark \end{array} \right]$$

optimize via substitution

$$\begin{aligned} V(C_1, C_2) &= V(C_1) = \ln(C_1) + \ln((100 - C_1)(1.1)) \\ &= \ln(C_1) + \ln(100 - C_1) + \ln(1.1) \end{aligned}$$

$$\frac{\partial V}{\partial C_1} = \frac{1}{C_1} + \frac{1}{100 - C_1}(-1) = 0$$

$$\Rightarrow \frac{1}{C_1} = \frac{1}{100 - C_1} \quad C_2^* = (100 - 50)(1.1) = 55$$

$$\Rightarrow 100 - C_1 = C_1$$

$$\Rightarrow 100 = 2C_1$$

$$\Rightarrow C_1^* = 50$$

$$S^* = 100 - C_1^* = 100 - 50 = 50$$

what's the slope? (-1.1)

- (b) The government imposes a 20% tax on labor income. Solve for the new optimal levels of C_1 , C_2 , and S . Explain any differences between the new level of savings and the level in part (a), paying attention to any income and substitution effects.

what changes? only labor income changes... taxed @ 20%, thus

$$V(C_1, C_2) = V(C_1) = \ln(C_1) + \ln(\underbrace{100(1-0.2)}_{80} - C_1) + \ln(1.1)$$

$$\frac{\partial V}{\partial C_1} = \frac{1}{C_1} + \frac{1}{80 - C_1}(-1) = 0$$

$$\begin{aligned} C_2^* &= (80 - 40)(1.1) \\ &= 40(1.1) = 44 \end{aligned}$$

$$\Rightarrow \frac{1}{C_1} = \frac{1}{80 - C_1}$$

$$S^* = 80 - 40 = 40$$

$$\Rightarrow 80 - C_1 = C_1$$

$$\Rightarrow 80 = 2C_1$$

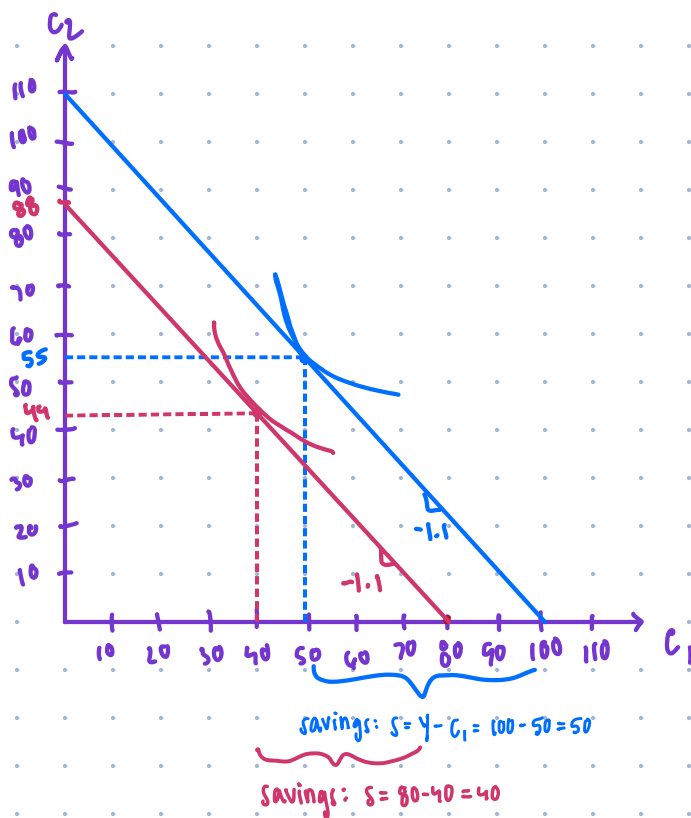
$$\Rightarrow C_1^* = 40$$

C_1^* , C_2^* , and S^* all decreased!

what is the new slope? still -1.1 (relative prices same)

what does that tell us about sub effect? there is NONE

this is a pure negative income effect



① $IBC: 100 = C_1 + \frac{C_2}{1.1}$

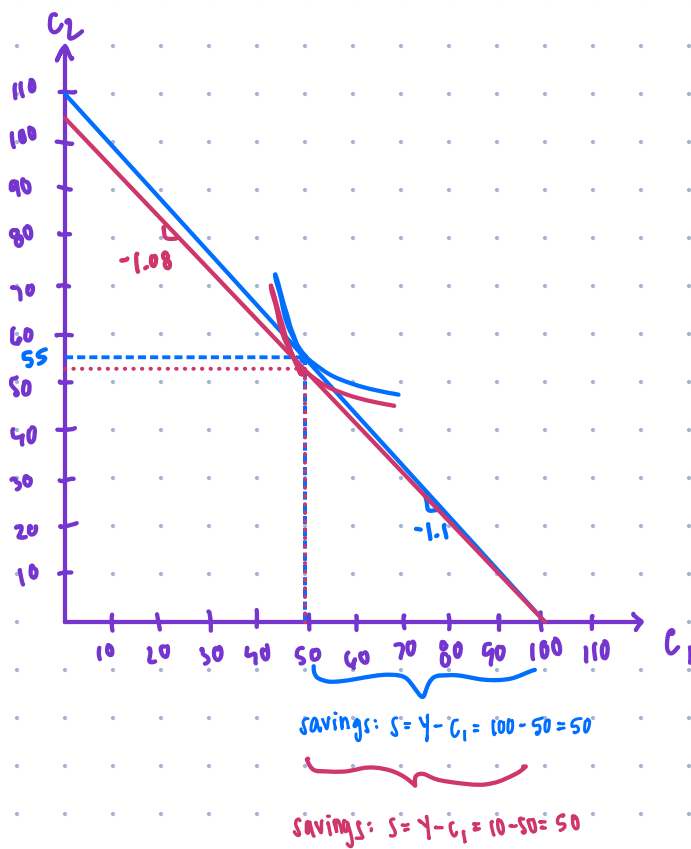
if $C_2 = 0$, $C_1 = 100$

if $C_1 = 0$, $C_2 = 100(1.1) = 110$

② $IBC: 80 = C_1 + \frac{C_2}{1.1}$

if $C_2 = 0$, $C_1 = 80$

if $C_1 = 0$, $C_2 = 80(1.1) = 88$



① same as before

② $Y = C_1 + \frac{C_2}{(1+r(1-\tau))}$

$100 = C_1 + \frac{C_2}{1.08}$

if $C_2 = 0$, $C_1 = 100$

if $C_1 = 0$, $C_2 = 100(1.08) = 108$

here, the relative prices of C_1 & C_2 changed due to lower returns on savings
in other words, sacrificing C_1 yields less increase in C_2 than before the tax!
so, substitution $C_2 \downarrow$, $C_1 \uparrow$
however, this is offset by the income effect, where $C_1 \downarrow$
 C_1 is the same w/ & w/o the tax

- (c) Instead of the labor income tax, the government imposes a 20% tax on interest income. Solve for the new optimal levels of C_1 , C_2 , and S . (Hint: What is the new after-tax interest rate?) Explain any differences between the new level of savings and the level in (a), paying attention to any income and substitution effects.

$$\begin{aligned} \text{so, now } C_2 &= S(1+r(1-\tau)) = (100-C_1)[1+0.1(1-0.2)] \\ &= (100-C_1)[1+0.08] \\ &= (100-C_1)(1.08) \end{aligned}$$

$$U(C_1, C_2) = U(C_1) = \ln(C_1) + \ln(100-C_1) + \ln(1.08)$$

$$\frac{\partial U}{\partial C_1} = \frac{1}{C_1} + \frac{1}{100-C_1}(-1) = 0$$

$$\Rightarrow \frac{1}{C_1} = \frac{1}{100-C_1} \quad \left\{ \begin{array}{l} 2C_1 = 100 \\ C_1^* = 50 \end{array} \right. \quad \left(\begin{array}{l} \text{got this above!} \\ S^* = Y - C_1 = 100 - 50 = 50 \end{array} \right)$$

$$\Rightarrow C_1 = 100 - C_1$$

$$\begin{aligned} \text{but } C_2^* &\text{ does change } (\downarrow) \\ C_2^* &= (100-C_1^*)(1.08) \\ &= 50(1.08) \\ &= 54 \end{aligned}$$

- (d) Returning to the labor income tax in part (a): What consumption tax rate would result in the same effects as the 20% labor income tax rate?

$$\text{Let's recall our IBC: } Y = C_1 + \frac{C_2}{(1+r(1-\tau))}$$

$$\text{w/o any tax: } 100 = C_1 + \frac{C_2}{1.1}$$

$$\text{w/ labor tax: } 100(1-0.2) = C_1 + \frac{C_2}{1.1} \Rightarrow 80 = C_1 + \frac{C_2}{1.1}$$

$$\text{w/ consumption tax: } 100 = (1+\tau)\left[C_1 + \frac{C_2}{1.1}\right]$$

[this taxes all consumption, so consumption in period 1 and period 2]

↓
you pay the tax for consuming.

$$80 = \frac{100}{1+\tau}$$

$$80(1+\tau) = 100$$

$$1+\tau = 1.25$$

$$\tau = 0.25$$

2 Business Taxation

2.1 Key Concepts

- Choice of entity: Business owners can choose between forming their business as a C-corporation, or as a pass-through (S-Corporations and Partnerships).
- C-corporations and pass-throughs are taxed in different ways:

	C-Corporation	Pass-through
Tax on \$1 of business income		
Tax on \$1 distributed to the owners		
Total after-tax income to owner		

2.2 Business Taxation and Investment

- An important question is how business taxation and the existence of deductions influences investment.
- In lecture, we assume a simple functional form based on Hall & Jorgenson (1967), where a firm earns profits as a function of capital, $F(K)$, and also faces a cost for using capital r , resulting in capital cost rK .
- Thus, the firm chooses capital to maximize $F(K) - rK$, yielding the first order condition $F'(K) = r$.
- A tax will distort the investment decision if it causes the firm to choose capital K such that $F'(K) \neq r$.

2.3 Practice Problem

- (a) An entrepreneur decides to start a new business. The gross profit function is $F(K) = pY - wL$ with $Y = L^{1/2}K^{1/2}$. Assume she'll hire just one employee and will have a fixed labor input $L = 1$ with $w = 2$. The market price is $p = 2$. Her cost of capital is $r = .25$.

Solve for her optimal investment K .

- (b) Is there any pure profit?

- (c) Let's say she forms the business as an S-corporation. The personal tax rate is $\tau_p = 25\%$, and *no deduction is allowed* for capital expenditures. Solve for her optimal investment K . Is there pure profit?

(d) Graph the two solutions (with x-axis K and y-axis rK) on the same graph.

(e) Now assume that a deduction is allowed for capital expenditures (i.e. that rK can be deducted from the firm's tax liability). What is the optimal K now?

3 Additional problems for practice

3.1 Gruber Ch. 22, Q.1

Suppose that a person lives for two periods, earning \$30,000 in income in period 1, which she consumes or saves for period 2. What is saved earns interest of 10% per year.

(a) Draw that person's intertemporal budget constraint.

(b) Draw that person's intertemporal budget constraint if the government taxes interest at the rate of 30%.

3.2 Gruber Ch. 22, Q.2

Suppose that the government increases its tax rate on interest earned. Afterward, savings increase. Which effect dominates, the income effect or the substitution effect? Explain.